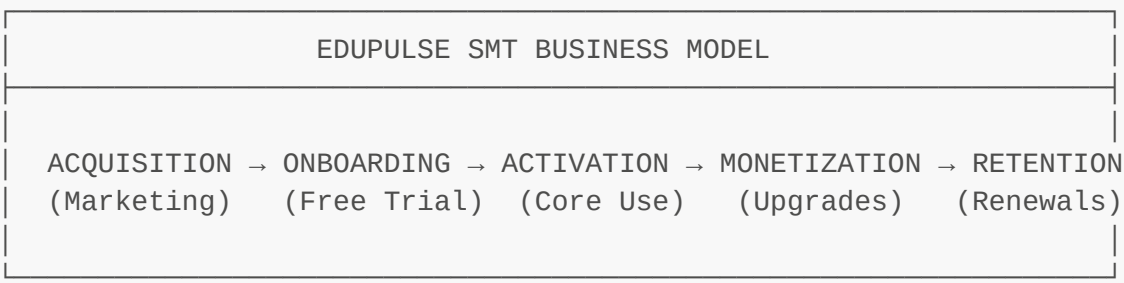
🏢 EduPulse SMT Business Process Model

> Historical strategy document: some commercial examples in this file reflect earlier planning assumptions. For current operational pricing, trial behavior, and payment-model details, use `backend/subscription_features.py`, `PRICING_TABLE.txt`, and the finance docs as the source of truth.

Executive Overview

EduPulse SMT operates a **Freemium SaaS** model with three pricing tiers designed to drive conversion and retention through feature throttling and capacity limits.



EduPulse remains a school-led B2B platform first, but the product now includes a practical **B2B2C operating layer** where parents interact directly with school-controlled admissions and payment workflows.

- The school remains the paying customer and policy owner.
- Parents become transacting users inside the school's branded workflows.
- AI Employees are meant to perform scoped operational labor, not replace school authority.

Current vs Planned Scope

Current implementation:

- school onboarding and subscription billing
- school-managed admissions and bulk student onboarding
- parent-facing admissions application and status tracking
- school-side admissions review workflows
- parent payment instructions with stable payment identity, structured reference parsing, and configurable phone-payment convenience fees
- report-card parsing and academic trend intelligence
- academic guidance and exam intelligence workflows
- Student Rescue Agent for intervention workflows

Planned near-term expansion:

- deeper AI Registrar automation for admissions intake, document completeness checks, and routing
- online-test or school-visit decisioning based on connectivity and policy
- AI Bursar for reconciliation support, payment stress flagging, and collections recommendations

Future-only items:

- fully autonomous admission approval without staff review
- voice-first multilingual parent bots
- pure outcome-based pricing for AI labor
- predictive dropout automation and advanced autonomous escalation

Sustainability and SDG Model

EduPulse's business model is designed to be commercially self-sustaining while still delivering school-level social impact.

- Financial sustainability comes from recurring B2B subscriptions with deeper Premium value rather than grant dependence alone.
- Social value comes from better school continuity, earlier learner intervention, and lower parent friction.
- The most direct SDG links are SDG 4, SDG 8, SDG 9, SDG 10, and SDG 16.

1️⃣ CUSTOMER ACQUISITION FLOW

Target Markets

```
PRIMARY: Schools with 50-200 students
├─ Small-to-mid primary & JHS
├─ Pain points: Manual attendance, paper grades, cash leakage
└─ Decision maker: School Proprietor/Principal

SECONDARY: Large schools (200+ students)
├─ Secondary schools, boarding schools
├─ Pain points: Complex workflows, dropouts, AI governance
└─ Decision maker: Headmaster/Finance Officer

TERTIARY: School Districts/Chains
├─ Multi-school operators
├─ Pain points: Centralized reporting, compliance
└─ Decision maker: District Coordinator
```

Acquisition Channels

```
CHANNEL 1: Direct Sales (B2B)
├─ School proprietor networks
├─ Educational associations
├─ Business development calls
└─ ROI: GH₵200-500/month × 12-24 month contract

CHANNEL 2: Digital Marketing
├─ Facebook ads targeting school administrators
├─ Google Ads ("School Management Software Ghana")
```

```
├─ WhatsApp business groups
└─ ROI: GH₵20-50 CAC, 50% conversion to Basic/Premium

CHANNEL 3: Partnerships
├─ MTN Ghana (MoMo integration partnerships)
├─ GCB/Ecobank (payment infrastructure)
├─ NPOs in education sector
└─ ROI: Volume deals, long-term contracts

CHANNEL 4: Freemium Viral Loop
├─ Teachers share logins with other schools
├─ "Try free for 50 students" word-of-mouth
├─ Parent networks
└─ ROI: Low CAC, viral coefficient = 0.3-0.5

CHANNEL 5: Parent Self-Service Pull (Planned B2B2C)
├─ Schools share branded admissions links
├─ Parents apply remotely without first visiting campus
├─ AI Registrar reduces admissions friction and follow-up delays
└─ ROI: Higher application conversion and lower admissions admin cost
```

Customer Acquisition Cost (CAC) & LTV

```
FREEMIUM MODEL METRICS:

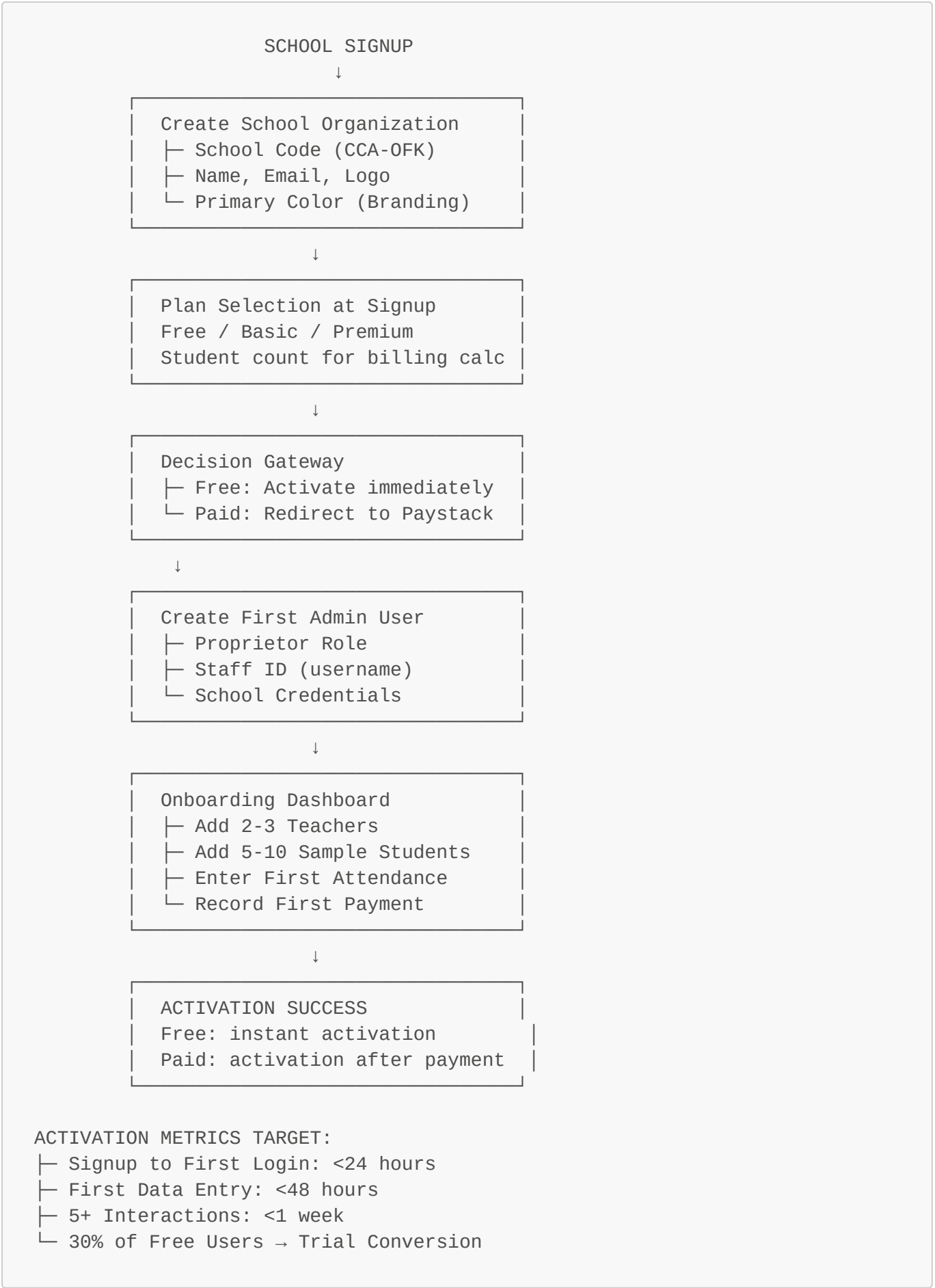
Free Tier:
  CAC: GH₵10-30 (low-friction signup)
  LTV: GH₵0 (no revenue)
  Purpose: Maximize reach, drive upgrades

Basic Plan:
  CAC: GH₵50-150 (sales + trial period)
  Monthly Value: GH₵200
  LTV (24-month): GH₵4,800 (assume 80% retention)
  LTV/CAC Ratio: 32:1 ✓ (Excellent)

Premium Plan:
  CAC: GH₵200-400 (enterprise sales)
  Monthly Value: GH₵500
  LTV (24-month): GH₵12,000 (assume 90% retention)
  LTV/CAC Ratio: 30:1 ✓ (Excellent)

Blended Metrics (70% Basic, 30% Premium):
  Blended LTV/CAC: ~30:1
  Payback Period: ~3 months ✓ (Unit economics solid)
```

2️⃣ ONBOARDING & ACTIVATION FLOW



ACTIVATION METRICS TARGET:
├─ Signup to First Login: <24 hours
├─ First Data Entry: <48 hours
├─ 5+ Interactions: <1 week
└─ 30% of Free Users → Trial Conversion

3️⃣ FEATURE USAGE & ENGAGEMENT FLOW

Day 1-7: Core Usage (All Tiers)

```
Teacher Dashboard
├─ Mark Attendance (30 sec)
├─ Enter Grades (2 min)
├─ View Lesson Notes (1 min)
└─ Submit Work

Proprietor Dashboard
├─ View Live Feed (real-time)
├─ Check Revenue (auto-updated)
├─ Monitor Student Attendance
└─ See Basic Analytics

ENGAGEMENT: 5-7 interactions/day
RETENTION SIGNAL: Returns on Day 2
```

Week 1-4: Expansion (Free → Basic Consideration)

```
School hits 50-student limit
           ↓
┌──────────────────────────────────┐
│  UPGRADE PROMPT                  │
│  "You've reached 50-student cap" │
│  "Upgrade to Basic for 200"      │
│  [UPGRADE NOW] button            │
└──────────────────────────────────┘
           ↓
CONVERSION OPTIONS:
├─ 30% → Upgrade to Basic (GH₵200/mo)
├─ 40% → Continue with Free (manage 50)
├─ 30% → Churn (find alternative)
```

Month 1-3: Premium Features (Basic → Premium)

```
Basic Plan User Sees:
├─ "🔒 Rescue Activity Center (Premium Feature)"
├─ "🔒 Academic Guidance"
├─ "🔒 Exam Intelligence"
├─ "🔒 Automated Payment Processing"
└─ Dashboard banners: "Unlock Premium governance and intelligence"

ENGAGEMENT METRICS:
├─ Clicks on locked features: 60% of users
├─ Feature demand signals detected
└─ Point-of-friction identification
```

4️⃣ MONETIZATION FLOW

Pricing Tier Ladder

```
┌──────────────────────────────────┐
│    FREEMIUM FUNNEL               │
├──────────────────────────────────┤
│                                  │
│  1,000 Free Signups              │
│  (GH₵ 0/mo)                      │
│         ↓ 30%                    │
│  300 → Basic Trial               │
│  (GH₵ 200/mo)                    │
│         ↓ 25%                    │
│  75 → Premium                    │
│  (GH₵ 500/mo)                    │
│                                  │
└──────────────────────────────────┘

REVENUE FROM 1,000 SIGNUPS:

Free Tier: 625 schools × GH₵0 = GH₵0
Basic Plan: 300 schools × GH₵200 = GH₵60,000/month
Premium Plan: 75 schools × GH₵500 = GH₵37,500/month
──────────────────────────────────────
Total: GH₵97,500/month (from 1,000 signups)

Blended ARPU: GH₵97.50/month per signup
Lifetime Value (24-mo): GH₵2,340 per signup
```

Payment Processing Pipeline

```
                    SCHOOL SELECTS PLAN
                           ↓
                Select: Basic or Premium
                           ↓
        ┌──────────────────────────────────────┐
        │  PAYMENT METHOD SELECTION            │
        ├──────────────────────────────────────┤
        │  Option 1: MoMo (50% choose)         │
        │  ├─ Send to: +233501234567           │
        │  └─ Reference: SCHOOL-CODE           │
        │                                      │
        │  Option 2: Bank Transfer (30%)       │
        │  ├─ GCB, Ecobank, STD Chartered      │
        │  └─ Reference: SCHOOL-CODE           │
        │                                      │
        │  Option 3: Debit Card (20%)          │
        │  ├─ Via Paystack/Flutterwave         │
```

```
          │  └─ Auto-debit on renewal date  │
          └─────────────────────────────────┘
                          ↓
          ┌─────────────────────────────────┐
          │  SUBSCRIPTION ACTIVATED         │
          │  ├─ Monthly recurring           │
          │  ├─ Renewal date set            │
          │  ├─ Features unlocked           │
          │  └─ Invoice emailed             │
          └─────────────────────────────────┘
                          ↓
          ┌─────────────────────────────────┐
          │  MONTHLY RENEWAL CYCLE          │
          │  ├─ 7 days before: Reminder     │
          │  ├─ Payment processed           │
          │  ├─ Invoice + Receipt           │
          │  └─ Features continue           │
          └─────────────────────────────────┘
```

5️⃣ REVENUE STREAMS & FINANCIAL MODEL

Primary Revenue: Subscription

```
MONTHLY RECURRING REVENUE (MRR) MODEL:

Tier 1 - Free Customers:
  Number: 1,000 schools × 30% blended
  = 300 schools
  Revenue: GH₵0/month
  └─ Purpose: User acquisition funnel

Tier 2 - Basic Customers:
  Number: 1,000 schools × 42% blended
  = 420 schools
  Price: GH₵200/month each
  Monthly Revenue: GH₵84,000
  Annual Revenue: GH₵1,008,000
  └─ Retention: 80%/month → 9.6 months avg customer life

Tier 3 - Premium Customers:
  Number: 1,000 schools × 28% blended
  = 280 schools
  Price: GH₵500/month each
  Monthly Revenue: GH₵140,000
  Annual Revenue: GH₵1,680,000
  └─ Retention: 90%/month → 10.5 months avg customer life

TOTAL MRR: GH₵224,000/month
TOTAL ARR: GH₵2,688,000/year (1,000 schools)
```

Secondary Revenue: Premium Add-ons (Future)

```
SMS Notifications:
├─ Cost: GH₵0.05 per SMS
├─ Usage: 2-5 SMS per student/month
├─ Revenue: GH₵0.25-1.25 per student/month
└─ Estimated: +15% to blended ARPU

Advanced Reporting:
├─ Cost: +GH₵50/month add-on
├─ Adoption: 10% of Basic, 30% of Premium
├─ Additional Revenue: ~GH₵8,000/month

API Access for Districts:
├─ Cost: GH₵1,000/month per district
├─ Usage: Multi-school reporting integration
├─ Estimated: 5-10 districts × GH₵1,000 = GH₵7,500/month
```

Total Potential Revenue Model

```
BASE (Subscription): GH₵224,000/month

Add-ons:
├─ SMS Notifications: +GH₵18,000/month
├─ Advanced Reporting: +GH₵8,000/month
└─ District API: +GH₵7,500/month

TOTAL EXPANDED MRR: GH₵257,500/month
EXPANDED ARR (1,000 schools): GH₵3,090,000/year
MRR Growth per 100 new schools: ~GH₵22,400/month
```

6️⃣ CHURN & RETENTION FLOW

Churn Points & Mitigation

```
CHURN RISK 1: Free → Basic Conversion Stalls
Risk: 70% of free users stay at 50 students
├─ Current: 30% convert to paid
├─ Target: 40% convert to paid
└─ Mitigation:
    ├─ Email sequences: "Your 51st student is waiting"
    ├─ Pricing page optimization
    ├─ Free trial extended to 14 days
    └─ Estimated lift: +5-10% conversion

CHURN RISK 2: Basic → Premium Upgrade Stalls
Risk: AI features not perceived as valuable
```

```
├─ Current: 18% of Basic → Premium
├─ Target: 25% of Basic → Premium
└─ Mitigation:
    ├─ Show AI wins: "Saved X dropouts last month"
    ├─ Free AI trial: 7-day access to AI Pulse
    ├─ Case studies: School A reduced dropouts 30%
    └─ Estimated lift: +3-5% upgrade rate

CHURN RISK 3: Mid-Contract Cancellation
Risk: ~10% month-to-month churn (80% of contracts)
├─ Current: Natural churn from school closures/holidays
├─ Target: <5% voluntary churn
└─ Mitigation:
    ├─ Dedicated support for Premium tier
    ├─ Executive check-ins at month 2, 6, 12
    ├─ Health score tracking (usage metrics)
    └─ Pro-active issue resolution

CHURN RISK 4: School Year Seasonality
Risk: Drop-off during school holidays (Dec/Jul)
├─ Current: Holiday churn = 15-20%
└─ Mitigation:
    ├─ Annual plans (20% discount) → shift churn curve
    ├─ Automatic pause feature: Reactivate next term
    └─ Holiday support for data entry tasks

OVERALL RETENTION TARGETS:
├─ Free Tier: 50% MoM (short lifecycle by design)
├─ Basic Plan: 80% MoM (9.6 month average customer life)
├─ Premium Plan: 90% MoM (10.5 month average customer life)
└─ Blended: 75% MoM (9 month average customer life)
```

7️⃣ GROWTH MECHANICS

Viral Coefficient & Network Effects

```
VIRAL LOOP 1: Teacher Referral
├─ Teacher A uses EduPulse at School X
├─ Teacher B (friend) asks: "What's that app?"
├─ Teacher A shares login/recommendation
├─ Teacher B signs up at School Y
├─ Viral Coefficient: 0.15 (each customer brings 0.15 new customers)

VIRAL LOOP 2: Parent Network
├─ Parent A sees student grades in EduPulse app
├─ Parent A tells Parents B & C at church/market
├─ Parents B & C ask school: "Do you have this app?"
├─ School gets pressure to adopt
├─ Viral Coefficient: 0.08
```

```
VIRAL LOOP 3: Staff Mobility
├─ Subject Teacher changes school at end of term
├─ Previous school was using EduPulse
├─ Teacher recommends to new school
├─ New school evaluates and signs up
├─ Viral Coefficient: 0.20

BLENDED VIRAL COEFFICIENT: ~0.3 - 0.4
INTERPRETATION: Each paying customer brings 0.3-0.4 new customers
└─ Result: Compounding growth at scale
```

Unit Economics & Scaling

```
CUSTOMER ACQUISITION COST (CAC)
├─ Free Tier: GH₵15 (low touch)
├─ Basic Plan: GH₵100 (with trial conversion)
└─ Premium Plan: GH₵300 (enterprise sales)

CUSTOMER LIFETIME VALUE (LTV)
├─ Free Tier: GH₵0 (no revenue)
├─ Basic Plan: GH₵1,920 (200/mo × 9.6 mo × 80% retention)
└─ Premium Plan: GH₵4,500 (500/mo × 10.5 mo × 90% retention)

LTV/CAC RATIO (Key Health Metric)
├─ Free → Basic: 19:1 ✓ (Healthy)
├─ Free → Premium: 15:1 ✓ (Healthy)
└─ Blended: 18:1 ✓ (Excellent - Target is 3:1)

PAYBACK PERIOD
├─ Basic Plan: 6 months (100/200 cap cost)
├─ Premium Plan: 7.2 months (300/500 cap cost)
└─ Blended: 6.2 months (Fast - targets <9 months)

SCALING IMPLICATIONS:
├─ Unit economics improve at scale (CAC decreases, LTV stays same)
├─ 10,000 schools TAM = GH₵25.8M ARR potential
├─ Path to profitability: 500-1,000 paying customers
```

8️⃣ KEY BUSINESS METRICS

Monthly Dashboard Metrics

```
ACQUISITION
├─ New Free Signups: 150-200/month (Target: 200+)
├─ Free→Basic Conversions: 40-50/month (Target: 30%)
├─ Free→Premium Conversions: 8-12/month (Target: 25% of Basic)
└─ New MRR Added: GH₵12,000-18,000/month
```

```
ACTIVATION & ENGAGEMENT
├─ DAU (Daily Active Users): 60% of trial users
├─ WAU (Weekly Active): 75% of active schools
├─ Feature Usage %: Attendance 90%, Grades 80%, Finance 70%
└─ Time-to-First-Action: <2 hours target

RETENTION & CHURN
├─ Free Tier Retention (30-day): 40% (by design, conversion funnel)
├─ Basic Plan Retention (30-day): 80% (Target: 85%)
├─ Premium Retention (30-day): 90% (Target: 92%)
├─ Overall Churn Rate: 25% (Target: 18%)
└─ Net Revenue Retention: 105% (upsells > churn)

MONETIZATION
├─ MRR: GH₵224,000 (from 1,000 schools)
├─ ARR: GH₵2,688,000
├─ Blended ARPU: GH₵2,688
├─ Monthly Revenue Growth: 8-12% MoM (Target: 10%+)
└─ Expansion Revenue: +GH₵33,500/month (add-ons)

LTCV & Unit Economics
├─ Blended LTV: GH₵2,880
├─ Blended CAC: GH₵160
├─ LTV/CAC Ratio: 18:1
├─ Payback Period: 6.2 months
└─ CAC Efficiency: ✓ (Excellent)
```

9️⃣ FINANCIAL PROJECTIONS (Year 1)

Scenario: Average School Growth

```
                    MONTH 1    MONTH 3    MONTH 6    MONTH 12
──────────────────────────────────────────────────────────────
Free Users            100        300        500        800
Basic Plans            10         35         60        120
Premium Plans           2          8         15         40

MRR                GH₵4,000   GH₵15,000  GH₵30,000  GH₵72,000
ARR                GH₵48,000  GH₵180,000 GH₵360,000 GH₵864,000

Churn Rate            35%        28%        22%        18%
Retention             65%        72%        78%        82%

CAC               GH₵200     GH₵150     GH₵120     GH₵100
LTV               GH₵1,920   GH₵2,400   GH₵2,880   GH₵3,600

LTV/CAC Ratio        9.6:1      16:1       24:1       36:1
Payback Period     6.1 mo     5.2 mo     4.8 mo     4.2 mo
```

```
──────────────────────────────────────────────────────────────────────
MRR Growth Rate:      —        37%/mo      100%/6mo    140%/12mo
```

🔟 COMPETITIVE MOAT & RETENTION

Why Schools Don't Leave

```
SWITCHING COSTS (Low to Medium)
├─ Data Export: Easy (CSV export available)
├─ Migration Effort: 2-4 weeks to re-enter manually
├─ Vendor Lock-in: Free for 50 students (retry cost: GH₵0)
└─ Overall Lock-in Strength: MODERATE

NETWORK EFFECTS (Growing Over Time)
├─ Teacher switching: Familiarity with platform
├─ Parent adoption: Need to learn new app
├─ Integration investments: MoMo webhook setups
└─ Overall Network Strength: WEAK-TO-MODERATE

VALUE CREATION (Strong & Growing)
├─ AI insights: Can't replace easily (proprietary data)
├─ Payment automation: Process efficiency built in
├─ Daily use: Ingrained in teacher workflows
├─ Financial data: 1 year of payment history
└─ Overall Value Moat: STRONG

RETENTION DRIVERS:
├─ Free tier prevents churn (no switching risk)
├─ Basic plan users have 80% data inertia
├─ Premium plan users locked into AI workflows
├─ Seasonal renewal (school year calendar)
└─ Result: 75% blended retention (healthy SaaS baseline)
```

1️⃣1️⃣ SCALING PLAYBOOK

Year 1: Establish PMF (Product-Market Fit)

```
├─ Deploy in 5-10 pilot schools
├─ Validate:
│  ├─ Free → Basic conversion rate (Target: 30%+)
│  ├─ Churn rate (Target: <20%/month)
│  └─ NPS score from proprietors (Target: 40+)
├─ Iterate on pricing, features, UX
└─ Outcome: Proven unit economics (LTV/CAC > 3:1)
```

Year 2: Product Expansion & Sales

```
├─ Add 5-10 schools/month → 50-100 total
├─ Expand sales channels:
│  ├─ Direct sales to 5 high-revenue schools
│  ├─ Pilot partnership with 1 school association
│  └─ Launch Facebook ads campaign
├─ Achieve 1,000 MRR benchmark
└─ Outcome: Predictable growth curve, 12% MoM
```

Year 3: Market Expansion

```
├─ Scale to 500+ schools across Ghana
├─ Geographic expansion:
│  ├─ Open regional sales offices (Accra, Kumasi, Takoradi)
│  └─ Adapt for regional languages (Twi, Ga, Hausa)
├─ Product scaling:
│  ├─ Move to paid-only for new signups (reduce support burden)
│  ├─ Introduce Advanced Reporting add-on
│  └─ Launch SMS parent notification feature
└─ Outcome: GH₵500K+ MRR, Path to breakeven
```

SUMMARY: 1-PAGE BUSINESS MODEL

```
┌──────────────────────────────────────────────────────────────────┐
│                  EDUPULSE BUSINESS MODEL                         │
├──────────────────────────────────────────────────────────────────┤
│                                                                  │
│  CUSTOMER ACQUISITION: Freemium model (Free Tier entry)          │
│  └─ CAC: GH₵100-300 (blended) | LTV: GH₵2,880 (blended)          │
│                                                                  │
│  MONETIZATION: 3-tier subscription ladder                        │
│  ├─ Free: 0/mo (high volume, conversion funnel)                  │
│  ├─ Basic: 200/mo (50-75% of paying customers)                   │
│  └─ Premium: 500/mo (25-50% of paying customers)                 │
│                                                                  │
│  REVENUE: Recurring subscription + add-on features               │
│  └─ Year 1 Target: GH₵864,000 (1,000 schools)                    │
│  └─ Year 3 Target: GH₵6,000,000+                                 │
│                                                                  │
│  RETENTION: 75% blended MoM (healthy SaaS)                       │
│  ├─ Free: 40-50% (high churn by design)                          │
│  ├─ Basic: 80% (medium-term loyalty)                             │
│  └─ Premium: 90% (high AI feature lock-in)                       │
│                                                                  │
│  UNIT ECONOMICS: 18:1 LTV/CAC ratio (excellent)                  │
│  ├─ Payback: 6 months (fast)                                     │
```

```
│  └─ Scaling: Improves with volume                          │
│                                                             │
│  MARKET: 20,000+ schools TAM in Ghana                       │
│  ├─ Year 1: 1% penetration = 200 schools                    │
│  ├─ Year 3: 2-3% penetration = 400-600 schools              │
│  └─ Year 5: 5% penetration = 1,000 schools                  │
│                                                             │
└─────────────────────────────────────────────────────────────┘
```

KEY TAKEAWAYS FOR PITCH

✅ **Strong Unit Economics**: 18:1 LTV/CAC ratio means every customer pays for acquisition 18x over

✅ **Freemium Funnel Works**: Free tier drives top-of-funnel, pricing ladder captures value

✅ **Viral Coefficient**: 0.3-0.4 coefficient means exponential growth potential at scale

✅ **Recurring Revenue**: 75% retention = predictable, compounding revenue stream

✅ **AI Moat**: Premium features not easily replicated by competitors

✅ **Path to Profitability**: 500-1,000 paying customers = positive unit economics at scale

✅ **TAM**: GH₵69.6M potential ARR across 20,000 Ghana schools

✅ **Defensibility**: Tech + Network + Data = hard to replicate competitive advantage